Jichen YAO

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EDUCATION

Northeastern University *Sep 2023 – Jul 2027 (Expected)*
Bachelor's Degree *Liaoning, China*

- Major: Computer Science and Technology | Average Score: 89/100
- Core Courses: Fundamentals of Programming (C Language), Object-Oriented Programming (C++), Intelligent Electronics and Information Technology, Algorithm Design and Analysis, Numerical Analysis, Digital Logic and Digital Systems, Basic Training of Information Technology, Data Structures, Probability Theory and Mathematical Statistics, Linear Algebra, Advanced Mathematics, Discrete Mathematics

RESEARCH EXPERIENC

Soul-Agent: An Architecture for Long-term Agent Identity and Evolution *Jun 2026 – Present*
Advisor: Assoc. Prof. Shi Feng

- Designed an agent research prototype that structurally separates persistent identity, long-term experience, current interaction state, and external actions, decoupling LLM reasoning from persistent-state governance.
- Implemented memory admission, state evolution, proactive behavior, version auditing, and non-destructive rollback, allowing long-term experience to update protected states only under evidence and permission constraints.
- Built 552 automated regression tests (100% pass rate) and 29 model-in-the-loop adversarial tests (25/29 blocked; 86%), covering semantic privilege escalation, slow poisoning, and forged evidence chains.

Cognitive-Inspired Strategy Optimization for Emotional Support Dialogue (ESC) *Dec 2025 – May 2026*
Advisor: Assoc. Prof. Shi Feng

- Proposed CogCSO, a cognitive-structure-aware preference optimization framework for ESC, encoding five therapeutic pacing principles into contrastive data construction to mitigate premature problem-solving.
- Extended the CSO-style MCTS preference construction pipeline by introducing a Cognitive Alignment reward signal, encouraging validation-before-intervention pacing during response selection.
- Constructed dialogue preference pairs from reward-guided MCTS rollouts and fine-tuned LLM backbones with LoRA and DPO using the LLaMA-Factory framework.
- Designed a robust evaluation framework (point-wise, pairwise blind, statistical tests) to analyze automated metric instability. Identified and analyzed the Verbosity Tax, validating it through cross-model contrast (LLaMA vs. Qwen).

Aspect-aware Multi-head Cross-Attention for Aspect-based Sentiment Classification *Jul 2025 – Nov 2025*
Advisor: Assoc. Prof. Shi Feng

- Proposed AMCA, an end-to-end model leveraging learnable aspect embeddings and multi-head cross-attention to extract semantics dynamically without external NLP tools.
- Redefined ABSA task objectives by shifting the focus from marginal classification accuracy to aspect detection completeness.
- Achieved state-of-the-art performance in aspect detection tasks across multiple datasets, demonstrating significant advantages in identifying implicit aspects through independent model design and systematic empirical validation.
- **Publication**: **Jichen Yao**, Shi Feng. "Aspect-aware Multi-head Cross-Attention for Aspect-based Sentiment Classification." *Accepted by ICNC-FSKD 2026*.

EXTRACURRICULAR ACTIVITY

Dalian Municipal Culture and Tourism Bureau Warm Winter Volunteer Service *Jan 2025*
Role: Volunteer | Organizer: Dalian Municipal Culture and Tourism Bureau

- Provided on-site assistance and information services to returning travelers at major transportation hubs including airports and railway stations.
- Delivered customer-oriented solutions to address inquiries regarding transportation, luggage, and local tourism information.
- Collaborated with a team of volunteers to ensure smooth operation of winter travel assistance initiative.

TECHNICAL SKILLS

- Languages: Mandarin (Native), English (TOEFL iBT Home Edition: 105)
- Programming: Proficient in Python, C, C++; Familiar with Shell scripting
- AI/ML Stack: PyTorch, LLaMA-Factory, HuggingFace, MCTS, DPO, Parameter-Efficient Fine-Tuning (LoRA)
- Tools: Git, Docker, LaTeX, Visual Studio Code, Linux Environment